

Research Intelligence Report

Blood Group Classification Using Microscopic Blood Cell Images — CNN/Deep Learning

Prepared for Paper Writing | May 2026

PART 1 — AVAILABLE PUBLIC DATASETS

Dataset 1 — PBC Dataset (Barcelona) ★ PRIMARY RECOMMENDATION

- **Full name:** Peripheral Blood Cell (PBC) Dataset
 - **Source:** Hospital Clínic of Barcelona / CellaVision DM96 analyzer
 - **Published:** April 2020 — Data in Brief, Elsevier (DOI: 10.1016/j.dib.2020.105474)
 - **Authors:** Acevedo, Merino, Alférez, Molina, Boldú, Rodellar
 - **Download:** <https://data.mendeley.com/datasets/snkd93bnjr/1>
 - **Size:** 17,092 images of individual normal cells
 - **Image size:** 360×363 pixels, JPG
 - **Classes (8):** Neutrophils, Eosinophils, Basophils, Lymphocytes, Monocytes, Immature Granulocytes, Erythroblasts, Platelets
 - **Annotation:** Expert clinical pathologists
 - **Notes:** First large publicly available normal peripheral blood cell dataset. No direct blood group labels. Can be used for morphology classification + framed as preliminary pipeline for blood group work.
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Dataset 2 — BCCD Dataset (Blood Cell Count and Detection)

- **Source:** GitHub — Shenggan/BCCD_Dataset (2024)
 - **Link:** https://github.com/Shenggan/BCCD_Dataset
 - **Size:** ~364 annotated images, detection-focused
 - **Classes:** WBC, RBC, Platelets (bounding box annotations)
 - **Notes:** Small but widely used in detection papers. Good for object detection baselines.
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Dataset 3 — TXL-PBC (Curated Multi-Source Dataset)

- **Published:** Scientific Data, Nature (2025)
- **Built from:** BCCD + BCDD + PBC + Raabin-WBC (4 merged sources)
- **Size:** 1,260 images, 18,143 bounding box annotations

- **Classes:** WBC, RBC, Platelets
 - **Notes:** Re-annotated with YOLOv8n + manual review for consistency. Best for detection papers.
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Dataset 4 — Munich Leukemia Laboratory Dataset (MLL)

- **Published:** Scientific Data, Nature (November 2025)
 - **Source:** Munich Leukemia Laboratory (largest European hematology lab)
 - **Size:** >40,000 single-cell images
 - **Classes:** 18 cytomorphology classes
 - **Collection period:** 2021–2024
 - **Notes:** Very large, expert-annotated, newest available. Not directly blood-group-labeled but best for morphology diversity.
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Dataset 5 — Raabin-WBC Dataset

- **Type:** White Blood Cell images
 - **Available on:** HuggingFace and GitHub
 - **Notes:** Commonly used in WBC classification benchmarks.
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Dataset 6 — Blood Group / Agglutination Image Datasets (Kaggle)

- **Kaggle: "Blood Group Detection Using Image Processing"** Link: <https://www.kaggle.com/code/grohith/blood-group-detection-using-image-processing/data> (Uses Blood Cell Images dataset, no direct ABO labels)
 - **Kaggle: "Blood Group Classification Based on Fingerprint"** Link: <https://www.kaggle.com/datasets/abhiramshibaraya/blood-group-classification-based-on-fingerprint> (Fingerprint-based, not microscopic — useful for comparison framing only)
 - **Kaggle: "Blood Cells Image Dataset"** Link: <https://www.kaggle.com/datasets/unclesamulus/blood-cells-image-dataset>
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Dataset 7 — Custom ABO Dataset (from IEEE 2022 paper)

- **Paper:** "A Novel Approach of Classifying ABO Blood Group Image Dataset Using Deep Learning" — IEEE 2022
- **DOI:** <https://ieeexplore.ieee.org/document/9752278/>
- **Notes:** Authors created their OWN ABO-labeled dataset (not publicly released). Achieved 96.7% accuracy using custom CNN vs AlexNet vs LeNet5. This is the most directly relevant

prior work.

DATASET STRATEGY FOR YOUR PAPER

Since no large public ABO-labeled microscopic dataset exists, your paper's FRAMING should be:

"We propose a CNN-based pipeline for automated blood group classification using agglutination/blood smear images, evaluated on [dataset name], with the aim of supporting AI-assisted hematological diagnostics."

You may construct a small custom dataset OR use agglutination card images (reaction visible as clumping).

PART 2 — PRIOR RESEARCH SUMMARY (What has been done before)

Paper 1 ★ Most Directly Related

- **Title:** A Novel Approach of Classifying ABO Blood Group Image Dataset Using Deep Learning
 - **Published:** IEEE ICCCNT 2022
 - **DOI:** <https://ieeexplore.ieee.org/document/9752278/>
 - **Method:** Custom CNN architecture
 - **Dataset:** Self-created ABO image dataset
 - **Accuracy:** 96.7%
 - **Compared to:** AlexNet, LeNet5
 - **Gap/Weakness:** Dataset not publicly released; no Rh factor included; no explainability; no comparison with modern transfer learning (MobileNetV2, EfficientNet)
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Paper 2

- **Title:** Non-Invasive Blood Group Detection Using CNN
 - **Published:** IJTRD, May-Jun 2023 (Vol. 10, Issue 3)
 - **Method:** CNN
 - **Gap:** Non-invasive via fingerprint, not microscopic blood image
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Paper 3

- **Title:** Deep Learning and Image Processing Based Blood Group Detection
- **Published:** IJCTT 2025, Vol. 73, Issue 5

- **Method:** K-Means clustering + SVM, ANN, Genetic Algorithm variants
 - **Tools:** Python, TensorFlow, Tkinter GUI
 - **Gap:** Uses classical ML classifiers; no transfer learning; no XAI
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Paper 4

- **Title:** A Deep Learning-Based Approach for Blood Group Prediction Using Fingerprint Images
 - **Published:** IJSCI 2025
 - **Method:** AI + Image Processing — captures agglutination reaction images processed through segmentation → feature extraction → classification
 - **Stages:** Preprocessing → Segmentation → Feature Extraction → Classification
 - **Gap:** Focused on agglutination card images, not raw microscopic blood smear images
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Paper 5

- **Title:** An Explainable AI-Based Blood Cell Classification Using Optimized CNN
 - **Published:** Journal of Pathology Informatics, July 2024 (DOI: 10.1016/j.jpi.2024.100389)
 - **Authors:** Islam, Assaduzzaman, Hasan — Daffodil International University, Bangladesh
 - **Method:** Optimized custom CNN + Transfer Learning (InceptionV3, MobileNetV2, DenseNet201) + SHAP + LIME + Grad-CAM + Grad-CAM++
 - **Result:** 99.12% accuracy, 99% precision, F1-score 99%
 - **Gap:** Focused on WBC subtype classification, NOT ABO blood group classification
 - **Relevance:** Your paper can build on this XAI methodology and apply it to ABO classification
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Paper 6

- **Title:** A Lightweight Explainable Deep Learning for Blood Cell Classification
 - **Published:** CMES (Tech Science Press), November 2025
 - **Method:** Lightweight CNN + Grad-CAM + LIME + SHAP
 - **Result:** High accuracy on 12-class classification
 - **Gap:** Blood CELL type classification, not blood GROUP (ABO) classification
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Paper 7

- **Title:** Deep CNNs for Peripheral Blood Cell Classification

- **Published:** arXiv 2021 (Gavas, Olpadkar — IIIT Hyderabad)
 - **Method:** Benchmarked 27 CNN architectures on PBC dataset (Barcelona)
 - **Best result:** 99.51% accuracy (ensemble of top models)
 - **Gap:** Only cell type classification; no blood group labels; ensemble = heavy model
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Paper 8

- **Title:** Classification of WBCs from Blood Smear Imagery Using ML and DL — A Global Scoping Review
 - **Published:** PLOS ONE, June 2024 (DOI: 10.1371/journal.pone.0292026)
 - **Coverage:** 136 studies from 2006–2023, 26 countries
 - **Key finding:** 54.4% of studies use CNNs; shift from classical ML to DL is strong
 - **Gap noted in review:** No standardized benchmark, models not generalizable across populations
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Paper 9

- **Title:** ReRNet: A Deep Learning Network for Classifying Blood Cells
 - **Published:** PMC/SAGE, 2023
 - **Method:** Custom deep network; also references CNN-RNN combo (Liang et al.) achieving 90.79% with Xception backbone
 - **Gap:** Cell type only, no blood group
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Paper 10

- **Title:** Efficient Blood Cell Classification Using U-Net + Lightweight CNN
 - **Published:** Scientific Reports, December 2025
 - **Method:** U-Net segmentation → lightweight CNN classification
 - **Classes:** 9 types (RBC, WBC subtypes, Platelet)
 - **Gap:** No blood group labels; segmentation adds complexity
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Paper 11

- **Title:** A Deep Learning Approach to Prediction of Blood Group Antigens from Genomic Data
- **Published:** Transfusion (Wiley), September 2024 (DOI: 10.1111/trf.18013)
- **Method:** Deep learning from genomic/genetic sequence data

- **Gap:** Genome-based, NOT image-based; completely different modality
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PART 3 — RESEARCH GAPS YOU CAN EXPLOIT

These are weaknesses in existing research that YOUR paper can address:

Gap 1 — No publicly available ABO-labeled microscopic image dataset

Almost all blood group classification papers either use self-created unpublished datasets or fingerprint images. Your paper can acknowledge this and either:

- (a) Use agglutination card images (visible clumping = ABO determinable), or
- (b) Construct a small annotated dataset as a contribution.

Gap 2 — No explainability in blood group classification

Existing ABO classification papers (IEEE 2022) use plain CNN with no XAI. Adding Grad-CAM makes your paper significantly stronger. The XAI papers exist for blood CELL type, not blood GROUP.

Gap 3 — No comparison of modern lightweight architectures for ABO classification

The 2022 IEEE paper compared only against AlexNet and LeNet5 (both old). No paper has compared MobileNetV2 vs EfficientNetB0 vs custom CNN specifically for ABO classification.

Gap 4 — No Rh factor (positive/negative) classification combined with ABO

Most papers do A, B, AB, O only (4 classes). Extending to 8 classes (A+, A−, B+, B−, AB+, AB−, O+, O−) is an unexplored direction.

Gap 5 — Class imbalance not addressed in blood group datasets

Blood group distribution globally is unequal (O is most common, AB is rare). No existing paper addresses class imbalance using SMOTE or weighted loss for ABO classification.

Gap 6 — Population/geographic bias

Studies noted by reviewers: models trained on Egyptian or Indian population data may not generalize globally. Acknowledging this as a limitation and discussing it shows academic maturity.

Gap 7 — No standardized preprocessing pipeline for ABO image classification

No paper has defined a reusable, standardized preprocessing pipeline (CLAHE + denoising + augmentation) specifically for blood group image datasets.

PART 4 — ACCURACY BENCHMARKS TO BEAT OR REFERENCE

Paper	Method	Task	Accuracy
IEEE 2022 (most relevant)	Custom CNN	ABO blood group	96.7%
Islam et al. 2024	Optimized CNN + XAI	WBC classification	99.12%
Gavas et al. 2021	Ensemble of 27 CNNs	PBC cell type	99.51%
EfficientNet-B3 (2025)	Transfer learning	Leukemia ALL	94.30% F1
MobileNetV2 2024	Transfer learning	WBC	~98.58%
CNN-RNN Xception	Hybrid	4-class blood cell	90.79%
Custom CNN (malaria)	Shallow CNN	RBC malaria	~97%

Your target: Beat the 96.7% from the IEEE 2022 ABO paper using:

- Better preprocessing (CLAHE)
- Modern architecture (MobileNetV2 or EfficientNetB0)
- Grad-CAM explainability
- Proper train/val/test split with reported F1, precision, recall

PART 5 — RECOMMENDED CITATIONS FOR YOUR PAPER

Use these in your Literature Review section:

1. Acevedo et al. (2020) — PBC dataset paper — cite for dataset description
2. IEEE 2022 ABO paper — cite as primary prior work to compare against
3. Islam et al. (2024) J. Pathology Informatics — cite for XAI methodology
4. Asghar et al. (2024) PLOS ONE — cite for WBC scoping review (shows field maturity)
5. Gavvas & Olpadkar (2021) arXiv — cite for CNN benchmarks on blood cell images
6. Scientific Reports 2025 U-Net paper — cite for preprocessing approaches
7. Moslemi et al. (2024) Transfusion Wiley — cite to differentiate (genomic vs image-based)
8. ReRNet 2023 PMC — cite for CNN-RNN comparison baseline

PART 6 — YOUR PAPER'S POSITIONING STATEMENT

Use this framing in your Abstract and Introduction:

"While prior studies have demonstrated CNN-based classification of peripheral blood cell subtypes with high accuracy, the specific problem of automated ABO blood group classification from microscopic images remains underexplored, with limited publicly available labeled datasets and no work addressing explainability or comparison of modern lightweight architectures. This study proposes a CNN-based pipeline for ABO blood group classification incorporating optimized preprocessing, transfer learning via MobileNetV2/EfficientNet, and Grad-CAM-based explainability, providing a transparent, efficient, and reproducible framework for AI-assisted hematological diagnostics."

PART 7 — SUGGESTED PAPER STRUCTURE (IEEE Format)

1. **Abstract** (~250 words)
 2. **Introduction** — importance of blood typing, limitations of manual methods, AI motivation
 3. **Related Work** — cite 8-12 papers from Part 5 above; organize by: (a) blood cell classification, (b) blood group detection, (c) XAI in medical imaging
 4. **Dataset Description** — describe dataset used, class distribution, preprocessing steps
 5. **Proposed Methodology** — CNN architecture, preprocessing pipeline, augmentation strategy
 6. **Experimental Results** — accuracy, precision, recall, F1, confusion matrix, Grad-CAM visuals
 7. **Comparative Analysis** — your model vs AlexNet vs LeNet5 vs MobileNetV2 baseline
 8. **Discussion** — what works, what doesn't, why
 9. **Conclusion & Future Work** — Rh factor extension, real-time deployment, larger dataset
 10. **References** (10-20 IEEE-format citations)
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Report compiled from live web research — May 2026 Sources: IEEE Xplore, PubMed/PMC, arXiv, Nature Scientific Data, PLOS ONE, Kaggle, Mendeley Data